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Batch 15

Algorithm

1. Read `size`.
2. Create the variable-length integer array `arr[size]`.
3. Read all elements of the array.
4. Read the window size `k`.
5. Pass the address of `arr` to `first_negative()`.
6. In `first_negative()`:
   - If `k <= 0` or `k > size`, display `-1`.
   - Use the outer loop to select each sliding window.
   - Use the inner loop to examine elements of that window from left to right.
   - When the first negative element is found, display it and `break` from the inner loop.
   - If no negative element is found, display `0`.
7. Separate the results with spaces.

Test Case Table

Test	Array	k	Expected Output
1	`12 -1 -7 8 -15 30 16 28`	3	`-1 -1 -7 -15 -15 0`
2	`1 2 -3 4 5 -6`	2	`0 -3 -3 0 -6`
3	`1 2 3 4`	2	`0 0 0`
4	`1 2 3`	4	`-1`
5	`-5 -2 -8`	2	`-5 -2`
6	`5 -4 7`	1	`0 -4 0`
7	`1 2 3`	0	`-1`

C program

```
#include <stdio.h>
```

```
void first_negative(int *arr, int size, int k)
{
    int i, j;
    int found;

    /* Check whether k is valid */
    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    /* Consider every window of size k */
    for (i = 0; i <= size - k; i++)
    {
        found = 0;

        /* Search for the first negative number in the window */
        for (j = i; j < i + k; j++)
        {
            if (*(arr + j) < 0)
            {
                printf("%d", *(arr + j));
                found = 1;
                break;
            }
        }

        /* No negative number found in the window */
        if (found == 0)
        {
            printf("0");
        }

        /* Print space between results */
        if (i < size - k)
        {
            printf(" ");
        }
    }

    printf("\n");
}

int main()
{
    int size, i, k;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size\n");
        return 0;
    }

    int arr[size];
```

```
    printf("Enter %d array elements:\n", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter k: ");
    scanf("%d", &k);

    first_negative(arr, size, k);

    return 0;
}
```

Execution

